CONCEPTUAL BRIDGE

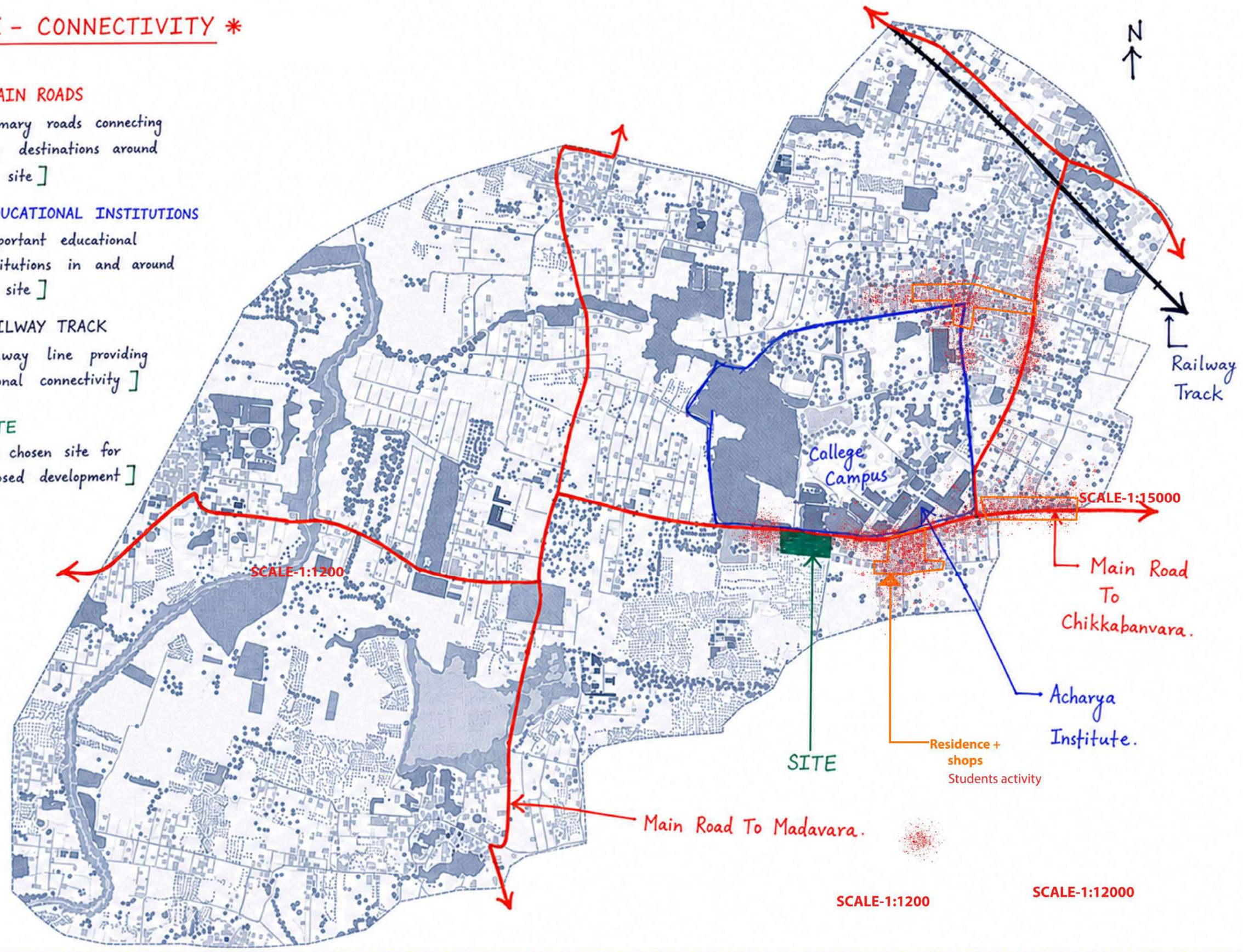
* SITE - CONNECTIVITY *

— MAIN ROADS
[Primary roads connecting key destinations around the site]

— EDUCATIONAL INSTITUTIONS
[Important educational institutions in and around the site]

— RAILWAY TRACK
[Railway line providing regional connectivity]

— SITE
[The chosen site for proposed development]



EXISTING URBAN ISSUES:



No integrated mixed-use environment to support daily student needs.

SCALE-1:1200



Street edges lack continuity and active pedestrian interaction.



Public spaces are unable to support increasing social interaction.



Traffic congestion increases during institutional peak hours.



EXISTING ISSUES:

PROPOSED MIXED-USE:



Development promotes walkability and reduces unnecessary travel.



Service access planned without disturbing residential activities.



Site connectivity improves relationship with surrounding institutions.



Creates active street frontage throughout the entire day.

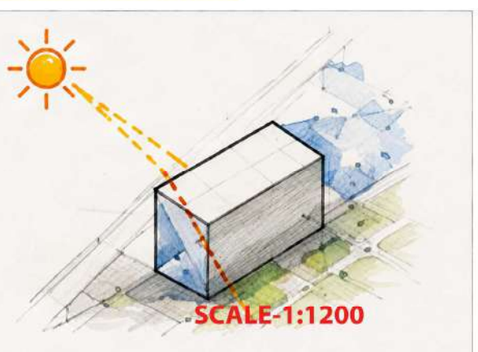


PLANING:

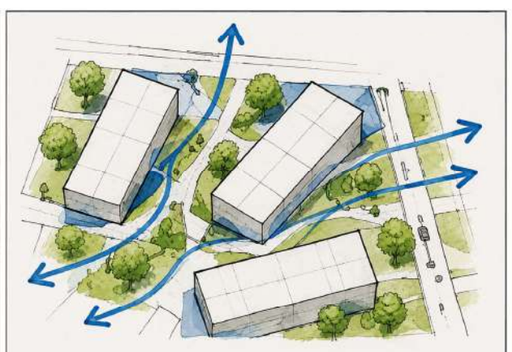
ENVIRONMENTAL AND DESIGN RESPONSE:



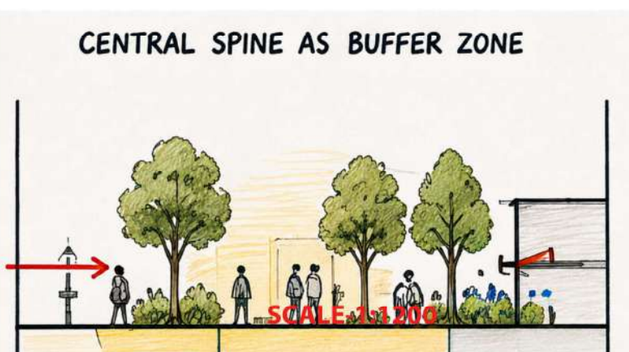
Open spaces designed to support shaded outdoor activities.



Reduced western exposure lowers internal heat accumulation.



Angled placement creates better airflow movement across the site.





OUTCOME:

8 HOUSE, COPENHAGEN, DENMARK

LOCATION:

PROJECT NAME:	8 HOUSE (8 TALLET)
ARCHITECTBIG :	BJARKE INGELS GROUP
LOCATION:	COPENHAGEN, DENMARK
YEAR COMPLETED:	2010
SITE AREA:	APPROX. 3.2 HECTARES
BUILT-UP AREA:	APPROX. 61,000 M²
BUILDING TYPE MIXED:	USE RESIDENTIAL DEVELOPMENT
STRUCTURE REINFORCED:	CONCRETE FRAME





INTRODUCTION:

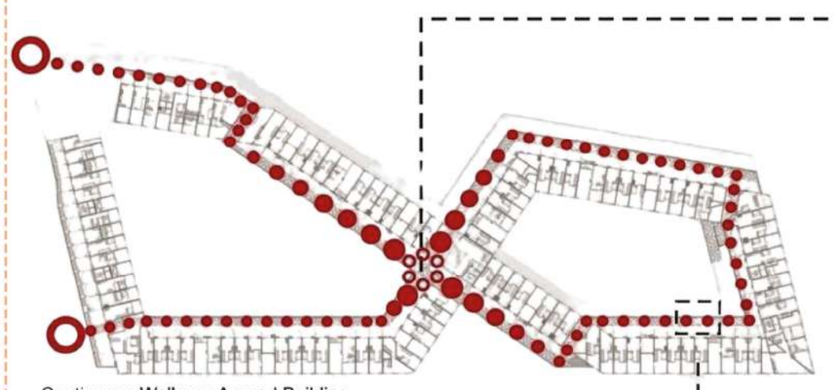
8 HOUSE IS A MIXED-USE RESIDENTIAL DEVELOPMENT DESIGNED AS A VERTICAL URBAN NEIGHBORHOOD INTEGRATING HOUSING, RETAIL, OFFICES, LANDSCAPE, AND PUBLIC CIRCULATION INTO ONE CONTINUOUS ARCHITECTURAL SYSTEM. THE PROJECT IS KNOWN FOR ITS INNOVATIVE RAMP CIRCULATION, COURTYARD PLANNING, AND COMMUNITY-ORIENTED SPATIAL DESIGN.



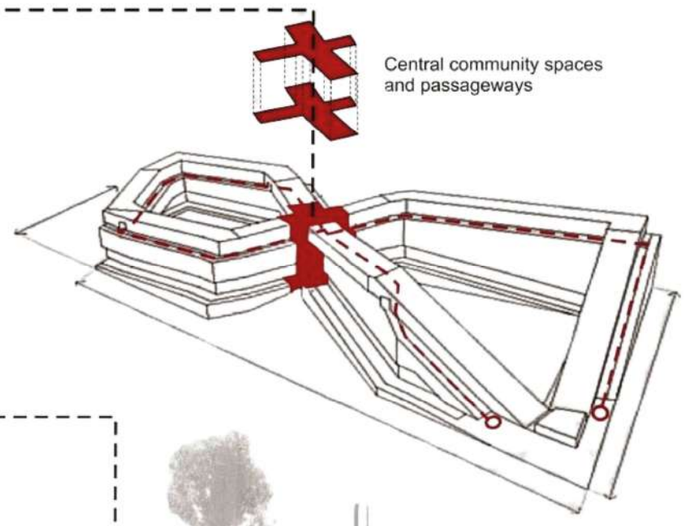


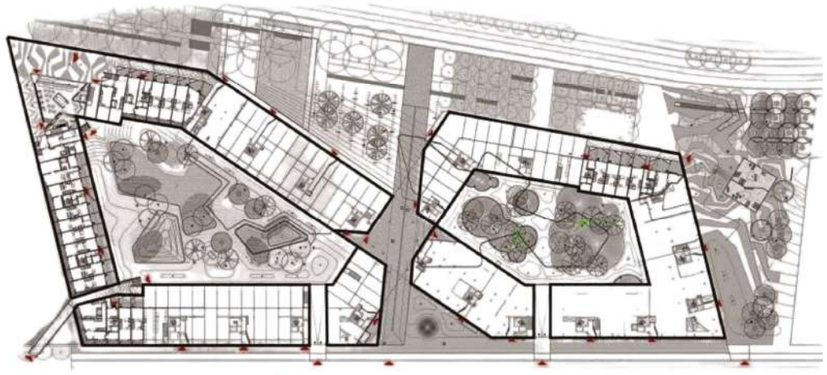
The **connection** between site and building **circulation**...



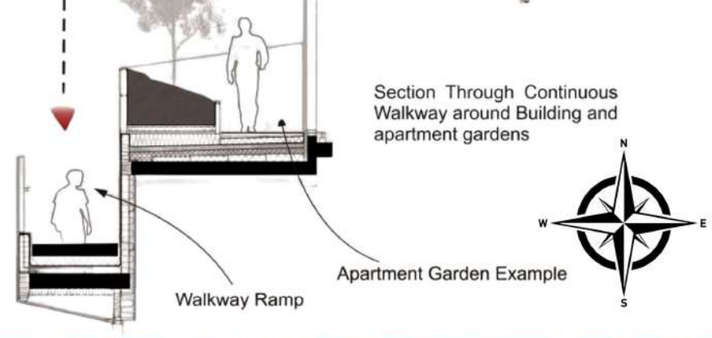


Continuous Walkway Around Building

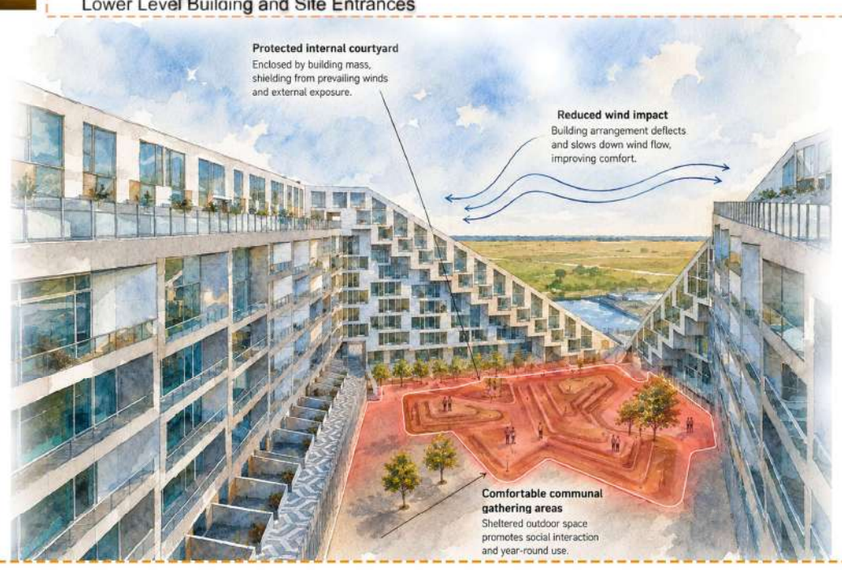




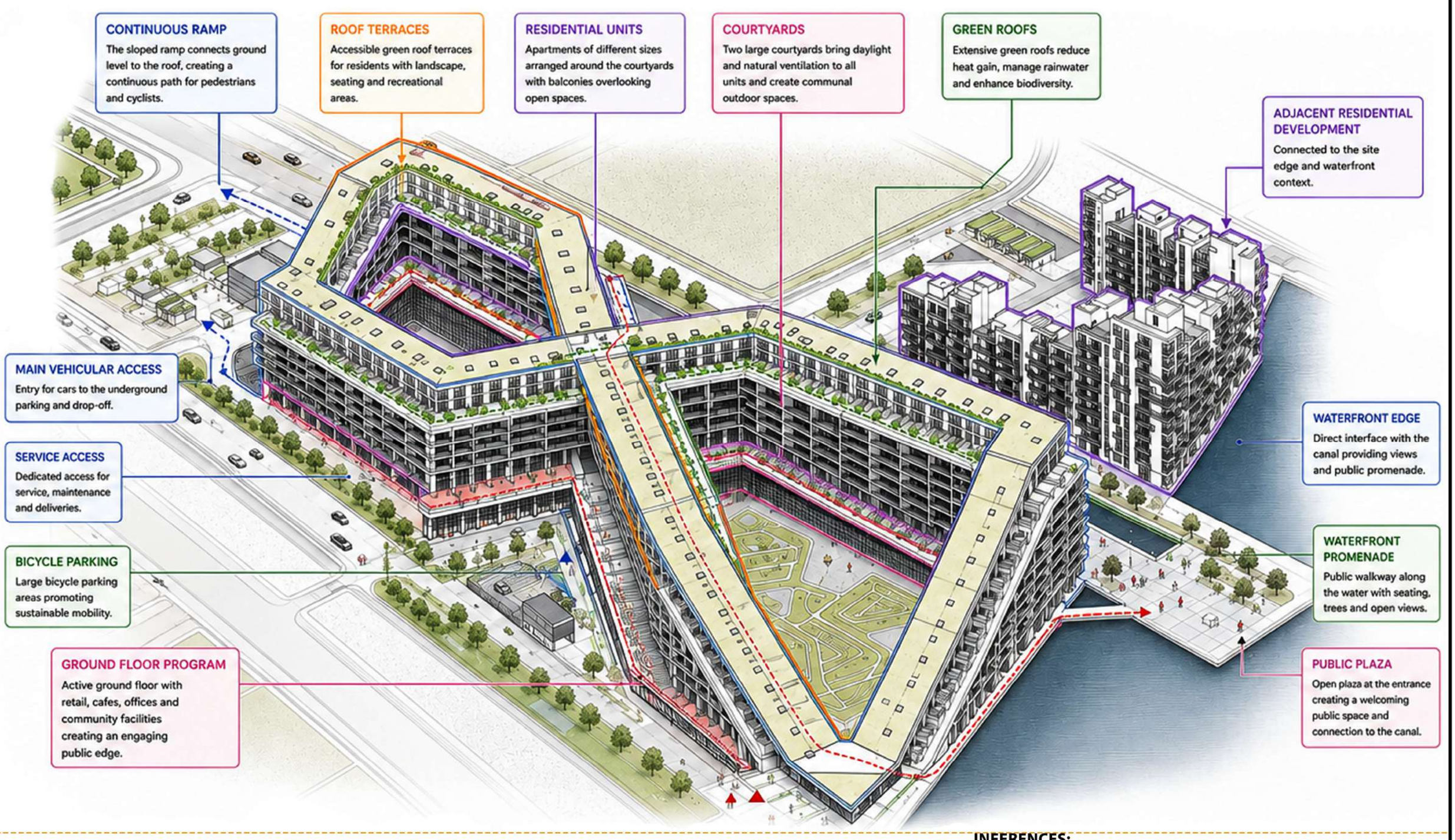
Lower Level Building and Site Entrances



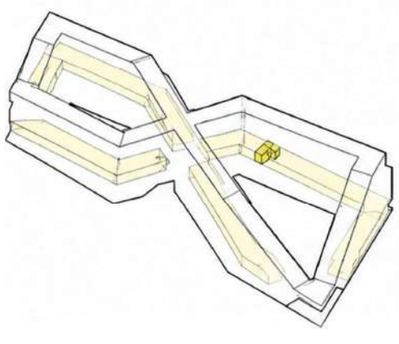




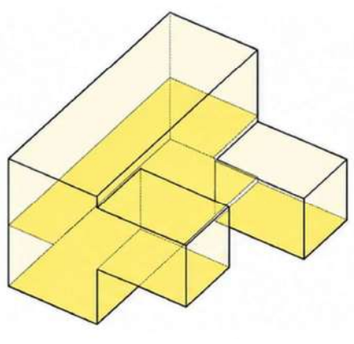




DUPLEX/STUDIO APARTMENT UNIT



LOCATION IN BUILDING



MASSING OF ONE UNIT



PLAN

SPACE AREA	APPROX.
LIVING AREA:	10–14 M²
BEDROOM ZONE:	8–10 M²
KITCHEN:	4–6 M²
DINING AREA:	4–5 M²
BATHROOM/WC:	2–3 M²
STAIRCASE:	3–5 M²
CIRCULATION:	2–4 M²
BALCONY/TERRACE:	5–8 M²

TOTAL BUILT-UP AREA:45–60 M² (APPROXIMATELY)

INFERENCES:

CONTINUOUS RAMP CIRCULATION CONNECTS RESIDENTIAL, COMMERCIAL, AND PUBLIC SPACES INTO ONE INTEGRATED URBAN SYSTEM.

INTERNAL COURTYARDS IMPROVE DAYLIGHT, VENTILATION, THERMAL COMFORT, AND PROTECTED COMMUNITY GATHERING ENVIRONMENTS.

TERRACED BUILDING FORM ALLOWS DEEPER NATURAL LIGHT PENETRATION INTO RESIDENTIAL UNITS AND CIRCULATION AREAS.

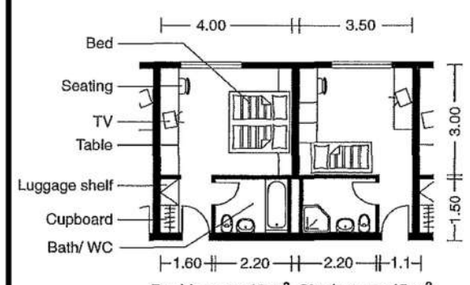
MIXED-USE ZONING ACTIVATES THE GROUND FLOOR THROUGH RETAIL, CAFÉS, AND PUBLIC INTERACTION SPACES.

ROOF TERRACES TRANSFORM UNUSED ROOF SURFACES INTO ACTIVE RECREATIONAL AND LANDSCAPE COMMUNITY SPACES.

EXTERNAL CIRCULATION IMPROVES VISUAL CONNECTIVITY AND ENCOURAGES SOCIAL INTERACTION BETWEEN RESIDENTS AND VISITORS.

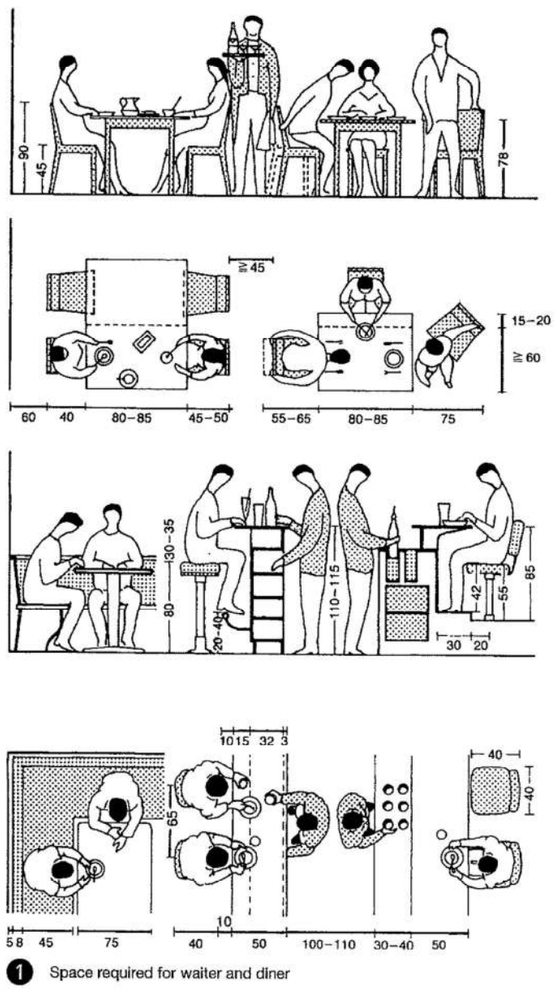
STANDARDS

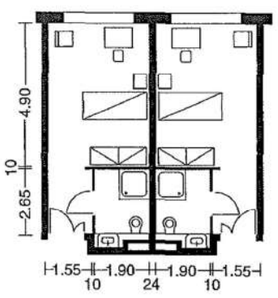
RESIDENTIAL + SERVICE:



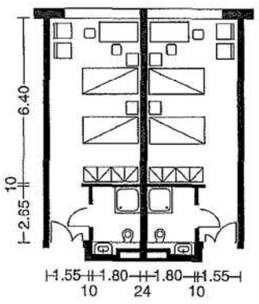
Double room 18 m² Single room 15 m²

3-star hotel rooms showing features and main dimensions (according to DEHOGA classification, single room slightly enlarged)



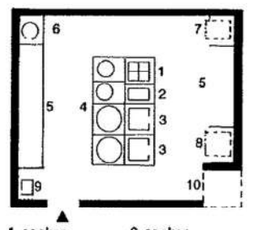


One-bed care room



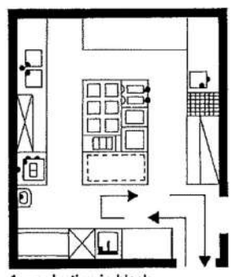
Two-bed care room

1 Space required for waiter and diner



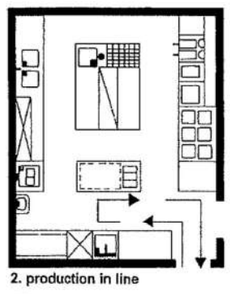
1 cooker
2 deep fat fryer
3 griddle
4 water boiler
5 work surface
6 cooker
7 double-deck oven
8 convectomat
9 hand basin
10 storage area

1 Basic organisation of the hot kitchen → 2 – 3



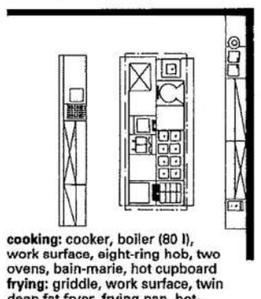
1. production in block

2 Kitchen for restaurant with 60–100 places



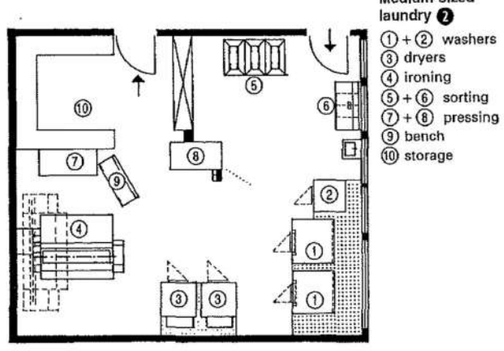
2. production in line

3 Kitchen for restaurant with 60–100 places



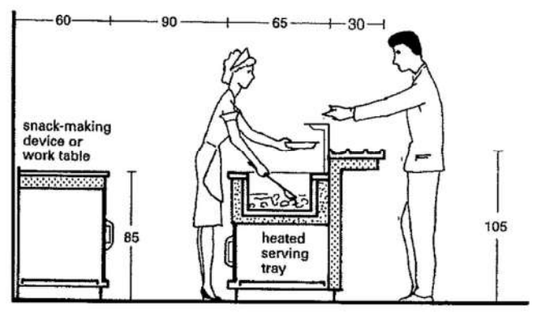
cooking: cooker, boiler (80 l), work surface, eight-ring hob, two ovens, bain-marie, hot cupboard
frying: griddle, work surface, twin deep fat fryer, frying pan, hot-air oven with table

4 Restaurant kitchen for 150–200 meals

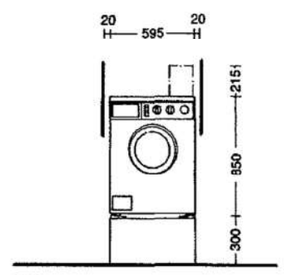


Medium-sized laundry 2
① + ② washers
③ dryers
④ ironing
⑤ + ⑥ sorting
⑦ + ⑧ pressing
⑨ bench
⑩ storage

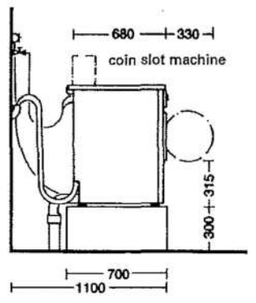
2 Medium-sized laundry



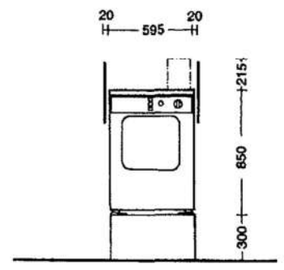
Food service, cafeteria



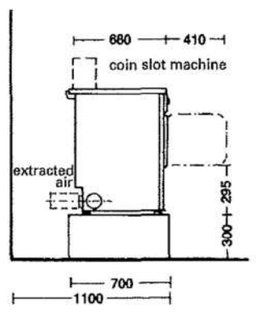
1 Automatic washer/spin dryer



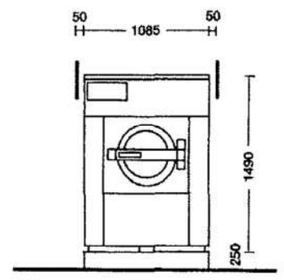
2 Side view → 1



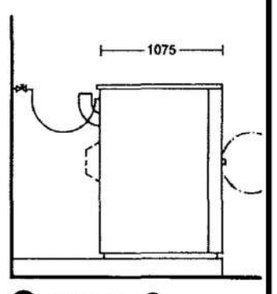
3 Automatic dryer



4 Side view → 3

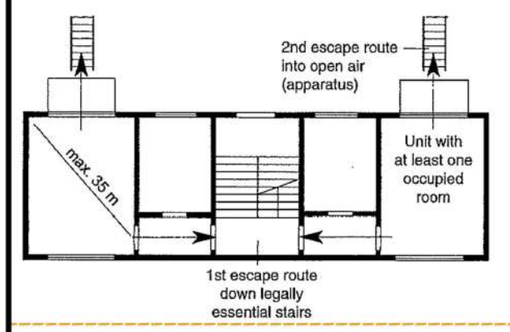


5 Automatic washer/dryer

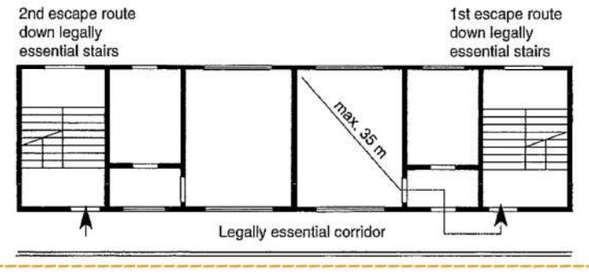


6 Side view → 5

Technical rules for fire protection (excerpt)



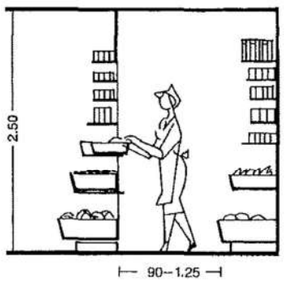
First and second escape routes for occupied units not on the ground floor (theoretical diagram, MBO)





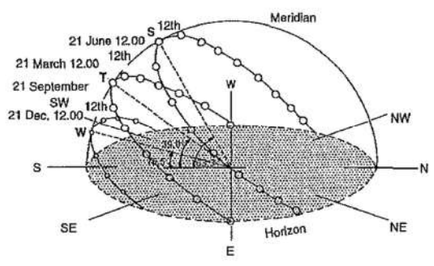
Wall shelves for bottles

Wall shelves for fruit and vegetables (goods to restock)

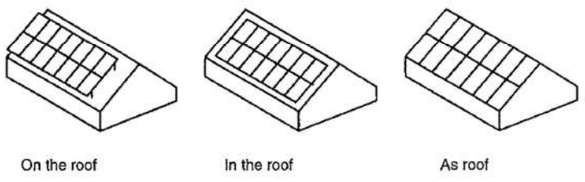


Wall with drawers, passage for restocking (sliding baskets are exchanged)

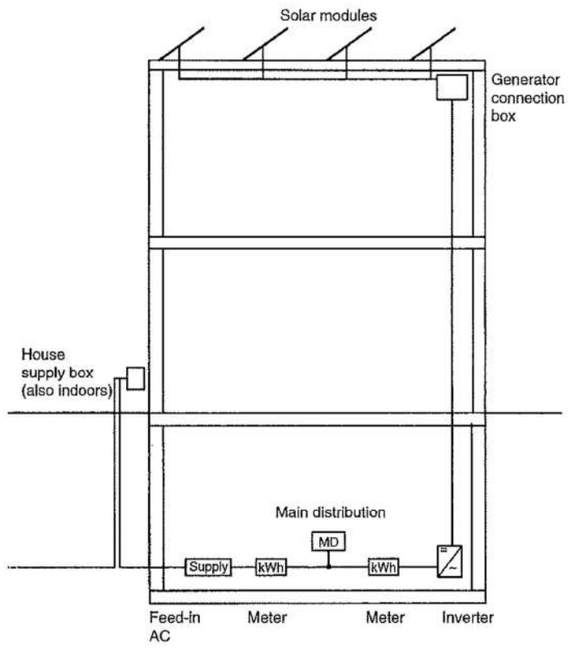
CLIMATE + DAYLIGHT



Path of the sun at the winter solstice = W, at the equinoxes = T, at the summer solstice = S, as experienced by a house or observer in north European countryside (51.5° latitude)



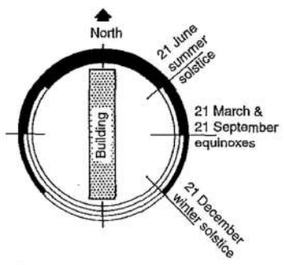
Possible roof-related arrangements of photovoltaic elements



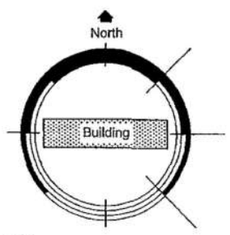
Principle of a grid-connected photovoltaic system (Bohne → refs)

DAYLIGHT
Physical basics
Position of the sun
Insolation
Shadow
Radiation energy
Window lighting
Rooflighting
Quality criteria
Directing sunlight
Shading

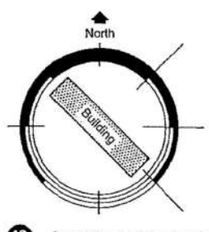
BS 8206-2
DIN 5034



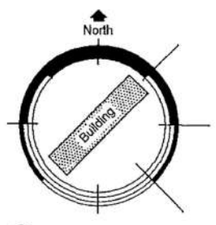
10 **A north–south block** receives sunshine on both sides, but has no windows on the north or south sides and therefore no sun into any rooms in November, December and January



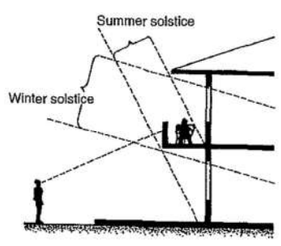
11 **An east-west block,** the best layout for small flats with 1–2 rooms: living rooms and bedrooms to the south (or open plan to the north); stairs, bathroom, lobby, kitchenette etc. to the north.



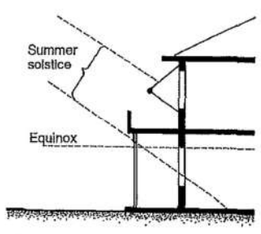
12 **A north-west to south-east block** is favourable for larger flats with bedrooms and utility rooms to the north-east and living rooms and children's rooms to the south-west



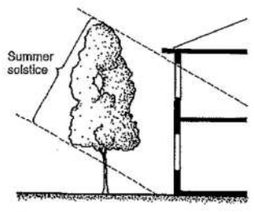
13 **A north-east to south-west block** is the best orientation for three- and four-room flats with living rooms and bedrooms to the south-east and utility and subsidiary rooms to the north-west



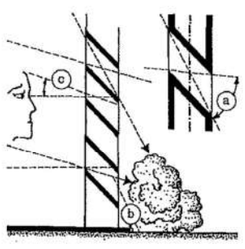
South side: sun and warmth penetrate deep into the house. In summer, sun and heat is kept out by windows and walls



15 **East side:** the flat sun rays from the east permit broad, mostly also wind-protected, terraces without hindering the entry of sunshine into the rooms

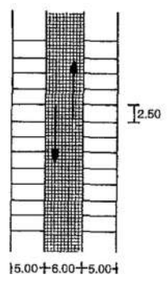


16 **West side:** disagreeable sunshine from the west and summer weather are best shaded by deciduous trees, which in the winter lose their leaves and permit the sun to enter

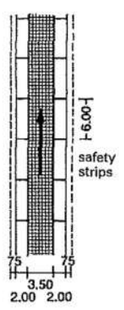


17 **A planked fence or balcony parapet** in this form lets most sunshine through → ⓐ, deflects the remaining rays → ⓑ and protects against overlooking and wind → ⓒ

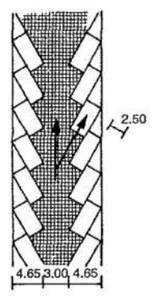
ROADS + PARKING



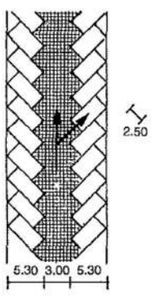
90° parking and exiting is possible in both directions. Access width 5.50 m for increased manoeuvring



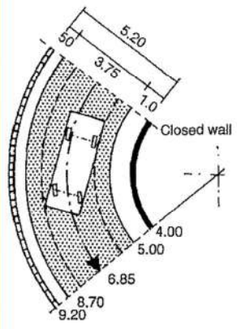
Parking parallel to the road



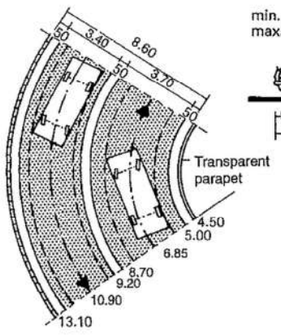
30° oblique parking and exiting is simple, but one-way traffic only



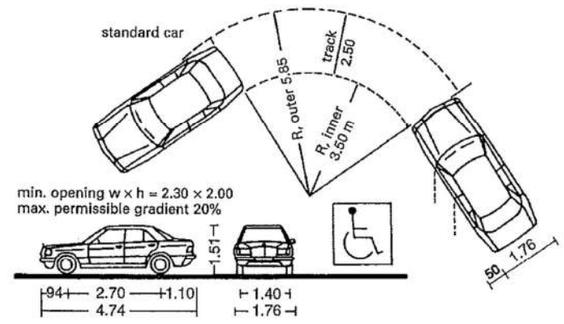
45° oblique parking, one-way traffic only



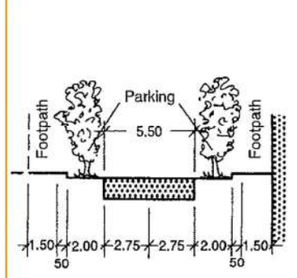
Minimum ramp widths



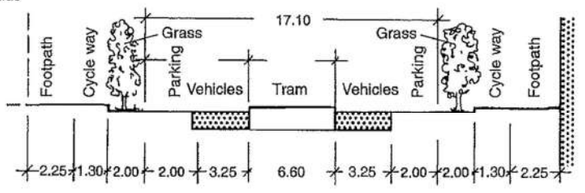
Minimum ramp widths in curve with minimum radius



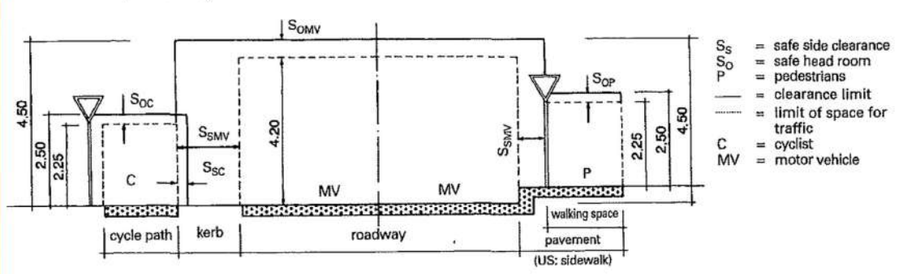
Standard car



Feeder road, carriageway designed for encounters of trucks with reduced speed; drivable side strip in case larger vehicles meet

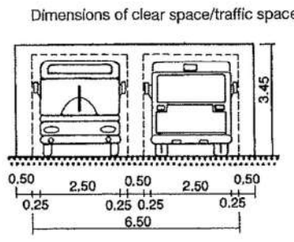


Pedestrian and cyclist area separated from road and parking by grass strips; tram on its own track bed

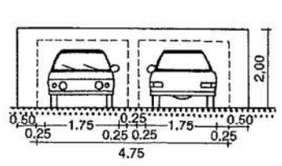


S_S = safe side clearance
S_O = safe head room
P = pedestrians
—— = clearance limit
····· = limit of space for traffic
C = cyclist
MV = motor vehicle

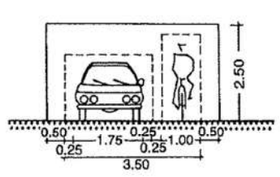
Dimensions of clear space/traffic space for vehicles



Bus/bus



Car/car



Car/bicycle

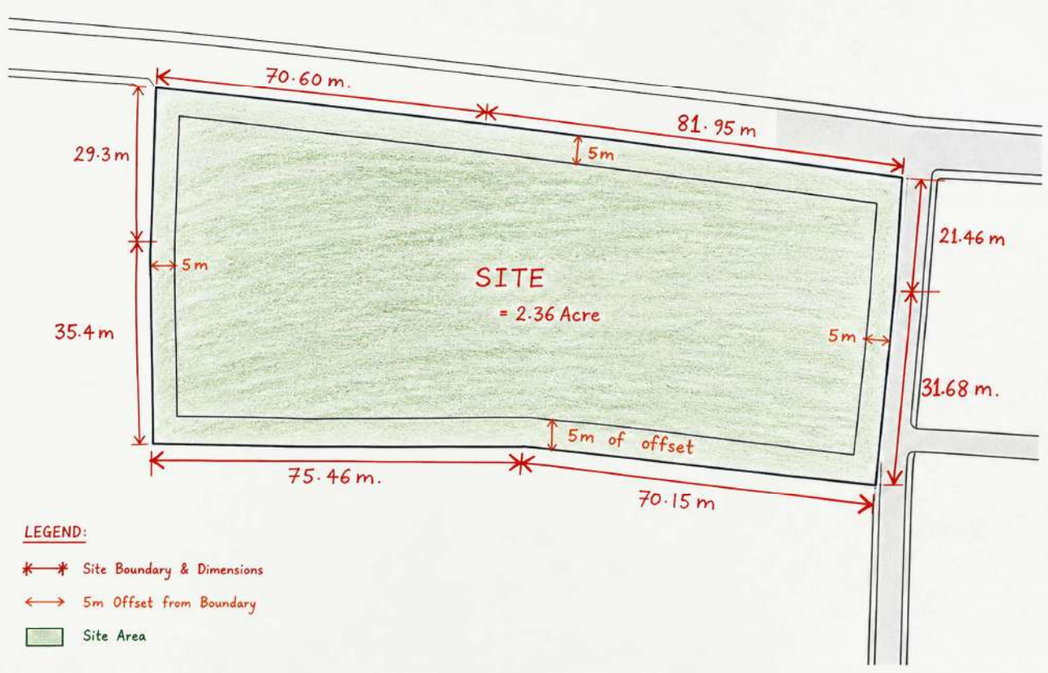
SITE ANALYSIS

SITE LOCATION:
THE SELECTED SITE IS LOCATED NEAR ACHARYA INSTITUTE WITHIN A DEVELOPING URBAN CORRIDOR. THE AREA IS CHARACTERIZED BY EDUCATIONAL INSTITUTIONS, RESIDENTIAL NEIGHBORHOODS, AND SUPPORTING COMMERCIAL ACTIVITIES.

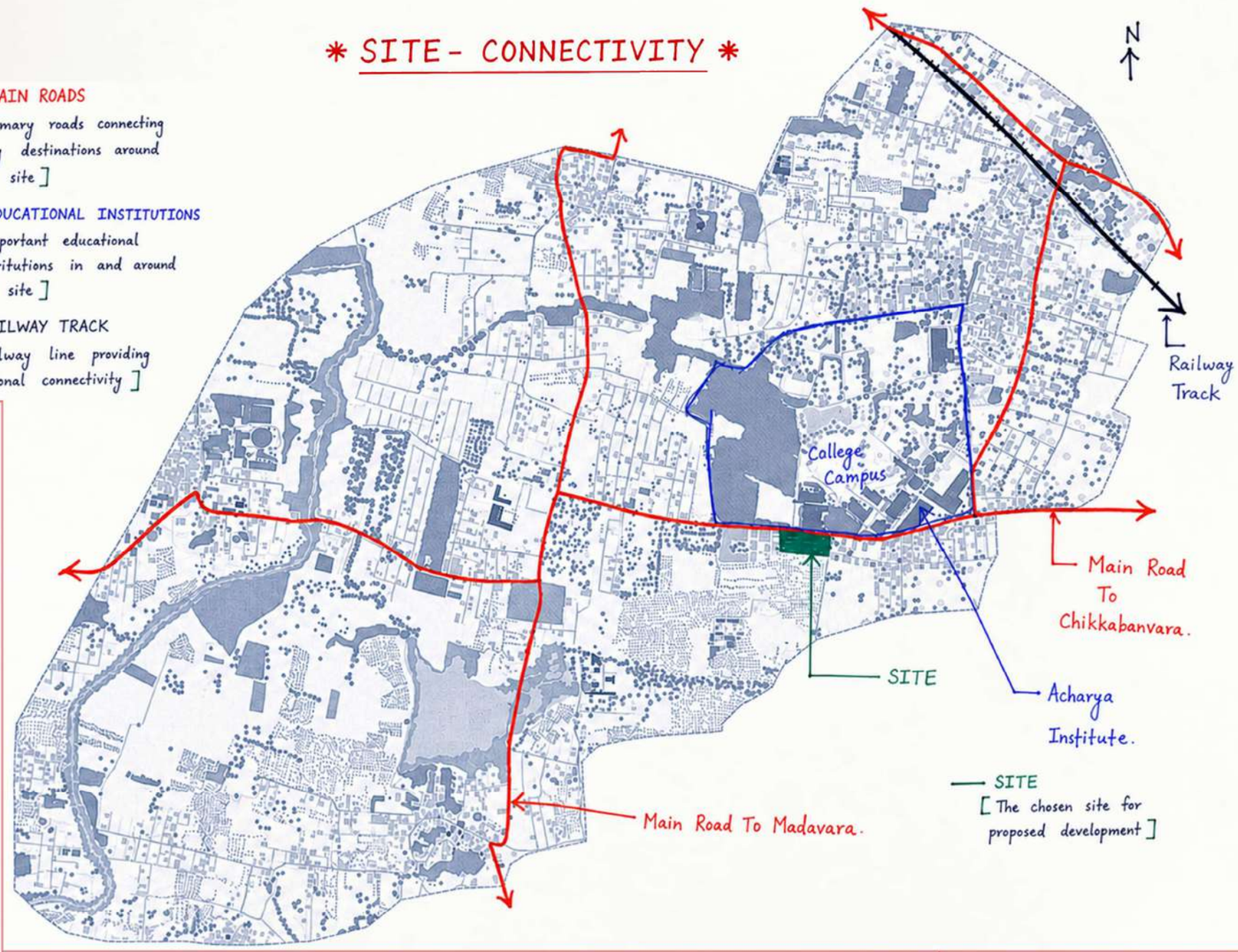
SITE AREA:
APPROXIMATELY 2.36 ACRES, WITH DIRECT ACCESS FROM PRIMARY AND SECONDARY ROADS.

CONTEXT CHARACTERISTICS:
THE SURROUNDING AREA CONSISTS OF INSTITUTIONAL BUILDINGS, OPEN LAND PARCELS, AND GROWING URBAN INFRASTRUCTURE SUPPORTING STUDENT POPULATION AND DAILY SERVICES.

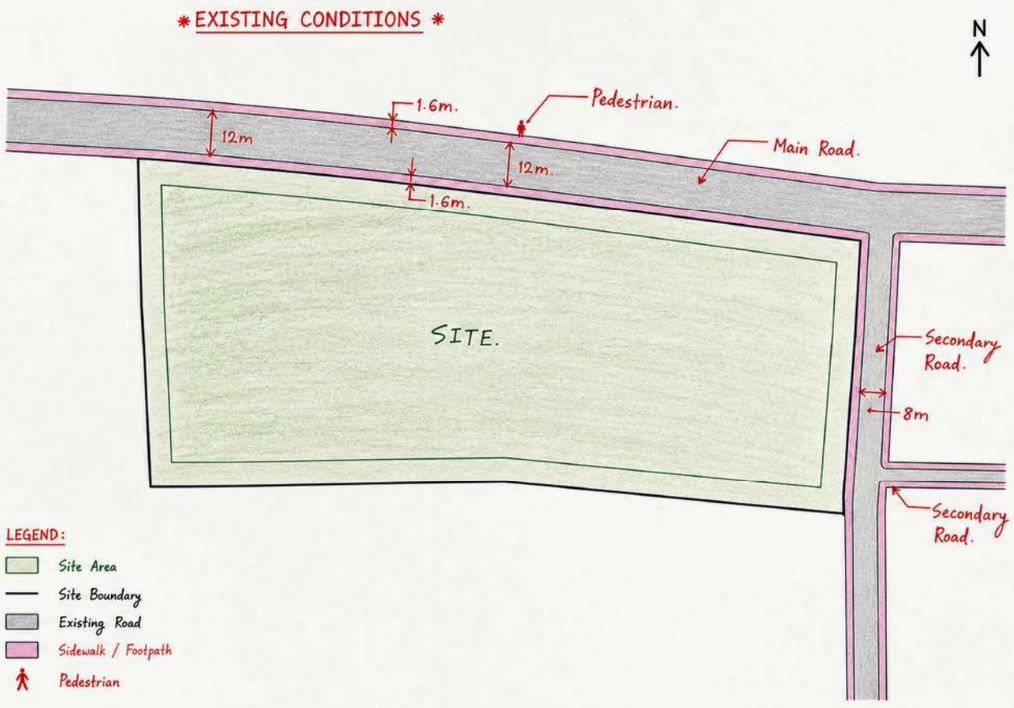
* SITE DIMENSIONDS *



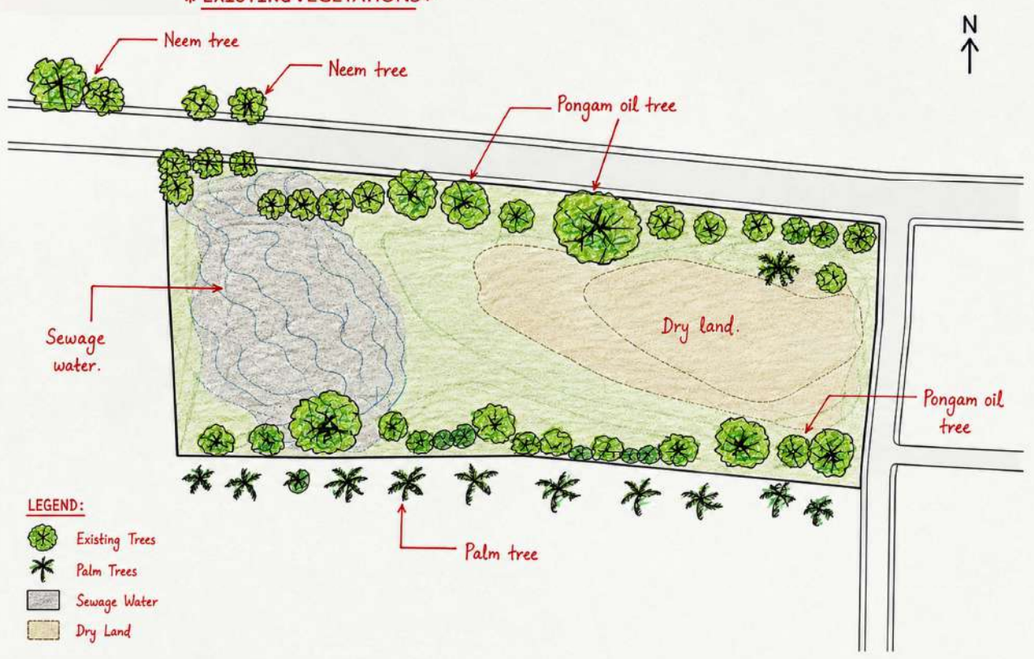
* SITE - CONNECTIVITY *



* EXISTING CONDITIONS *



* EXISTING VEGETATIONS *



SWOT ANALYSIS:

STRENGTHS:
SITE LOCATED NEAR MAJOR EDUCATIONAL INSTITUTIONS

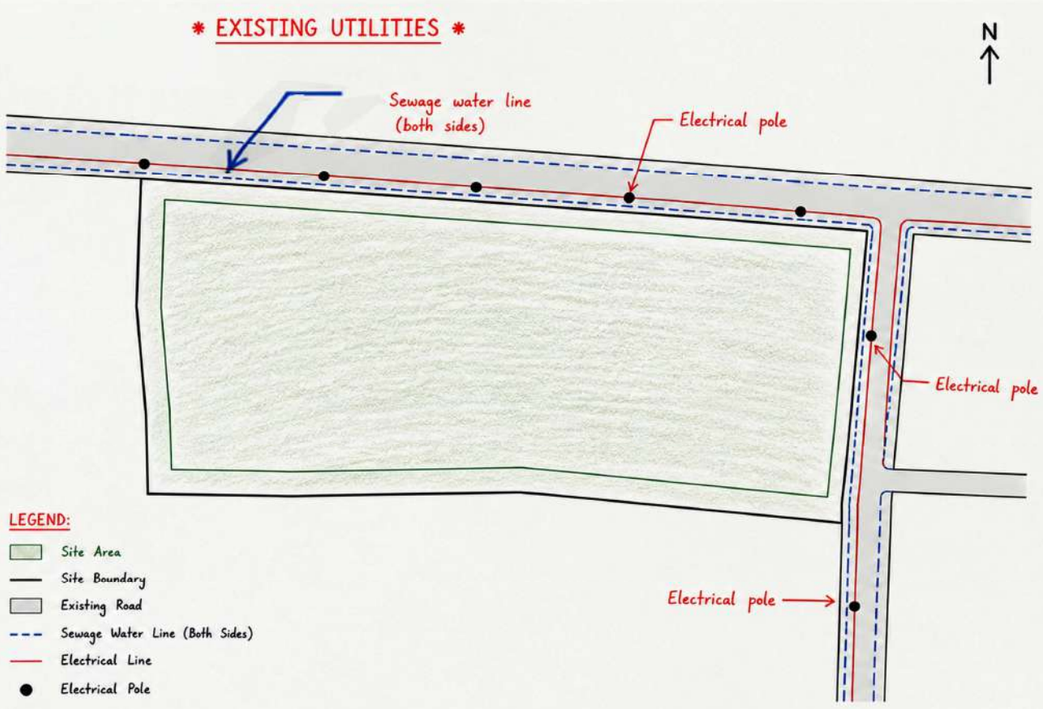
GOOD ROAD CONNECTIVITY AND ACCESSIBILITY

AVAILABLE INFRASTRUCTURE SERVICES NEARBY

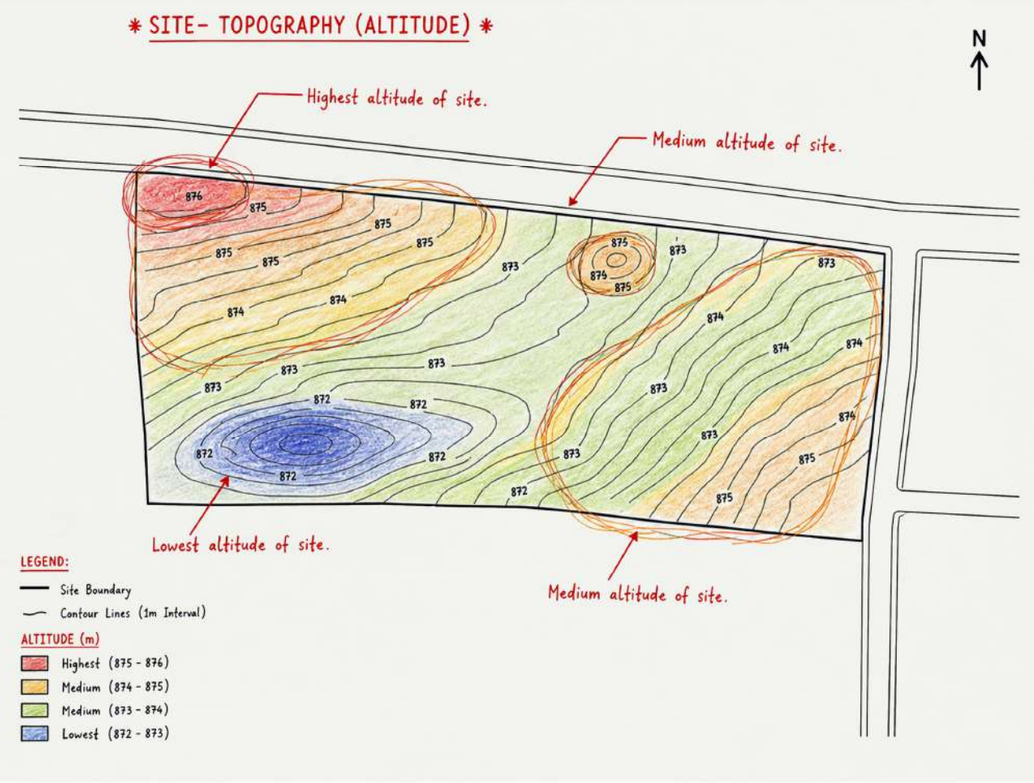
MODERATE SLOPE SUPPORTS NATURAL DRAINAGE

LARGE SITE AREA SUITABLE FOR MIXED-USE DEVELOPMENT

* EXISTING UTILITIES *



* SITE - TOPOGRAPHY (ALTITUDE) *



WEAKNESSES:
LOW-LYING AREA PRONE TO WATER ACCUMULATION

IRREGULAR SITE SHAPE REDUCES PLANNING EFFICIENCY

EXISTING SEWAGE WATER PRESENCE ON SITE

LIMITED PEDESTRIAN INFRASTRUCTURE

OPPORTUNITIES:
HIGH DEMAND FOR STUDENT HOUSING NEAR CAMPUS

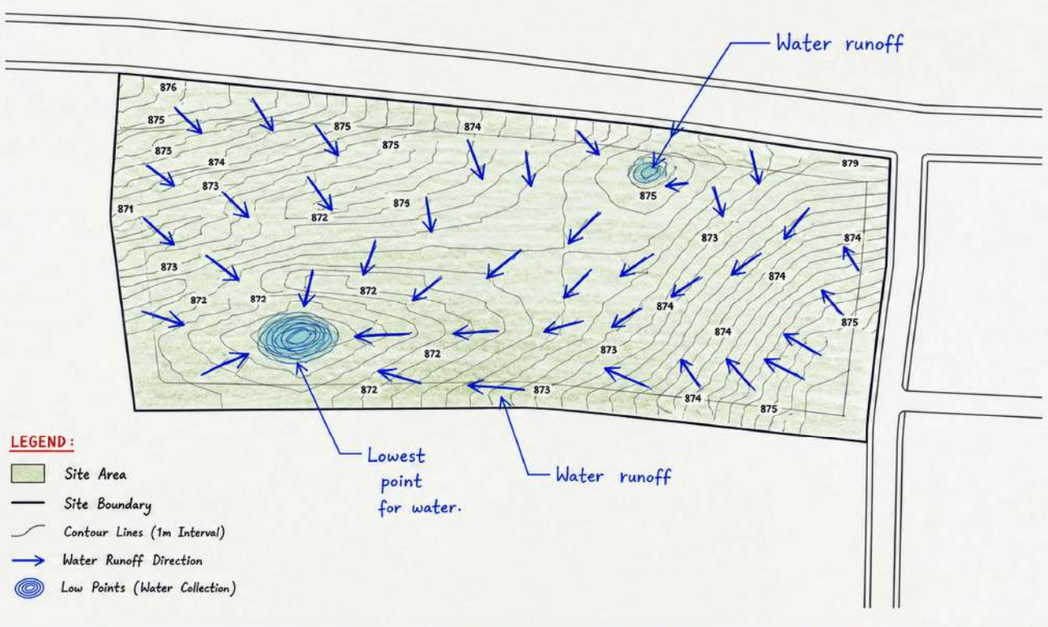
POTENTIAL INTEGRATION OF COMMERCIAL FACILITIES

RAINWATER HARVESTING AND SUSTAINABLE DRAINAGE

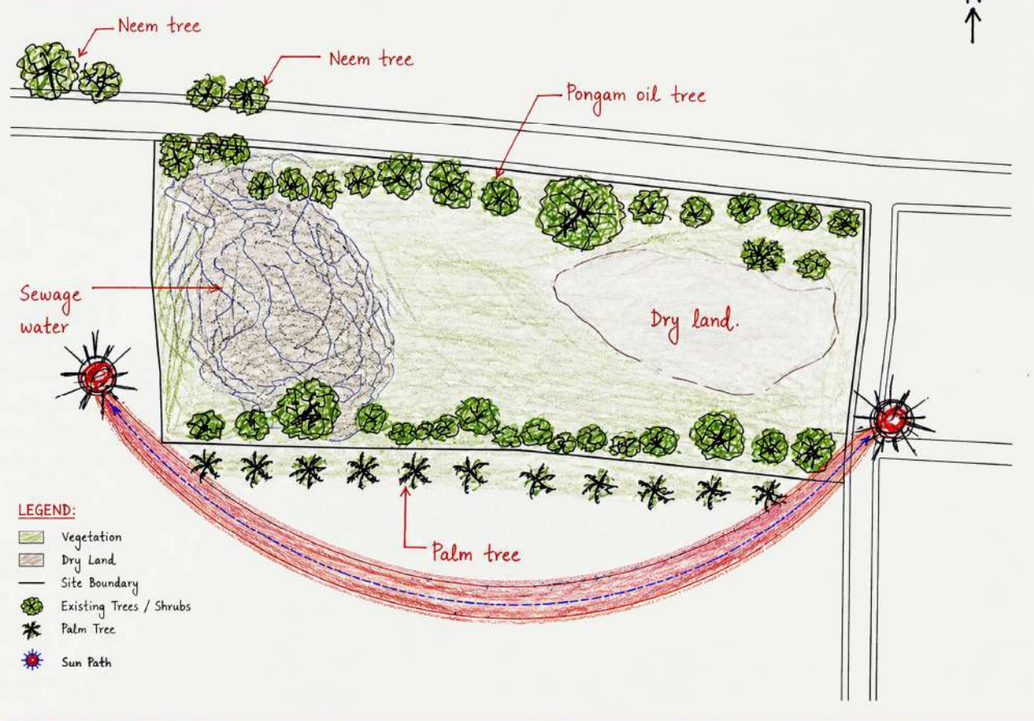
LANDSCAPE DEVELOPMENT USING EXISTING VEGETATION

FUTURE URBAN GROWTH AROUND THE CAMPUS

* TOPOGRAPHY & WATER RUNOFF *



* SITE ORIENTATION *



THREATS:
TRAFFIC CONGESTION FROM SURROUNDING ROADS

FLOODING RISK DURING HEAVY RAINFALL

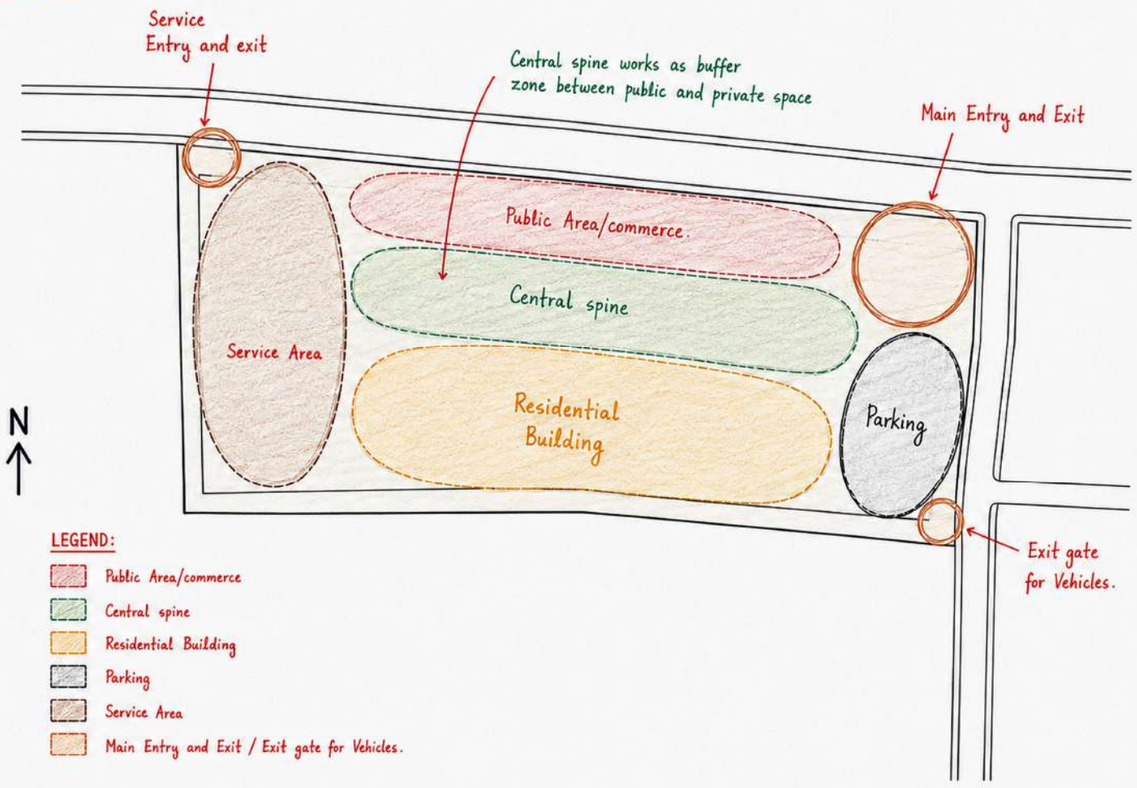
INFRASTRUCTURE PRESSURE DUE TO DEVELOPMENT

ENVIRONMENTAL IMPACT IF DRAINAGE UNMANAGED



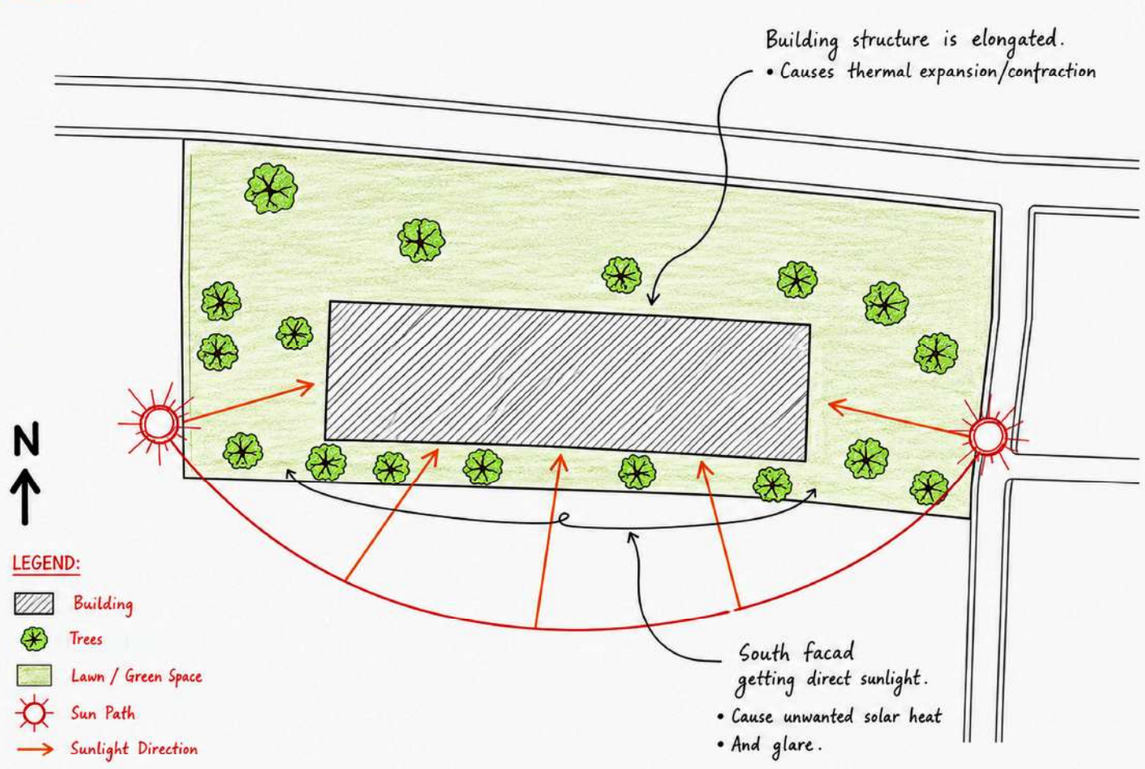
CONCEPT

STEP:1



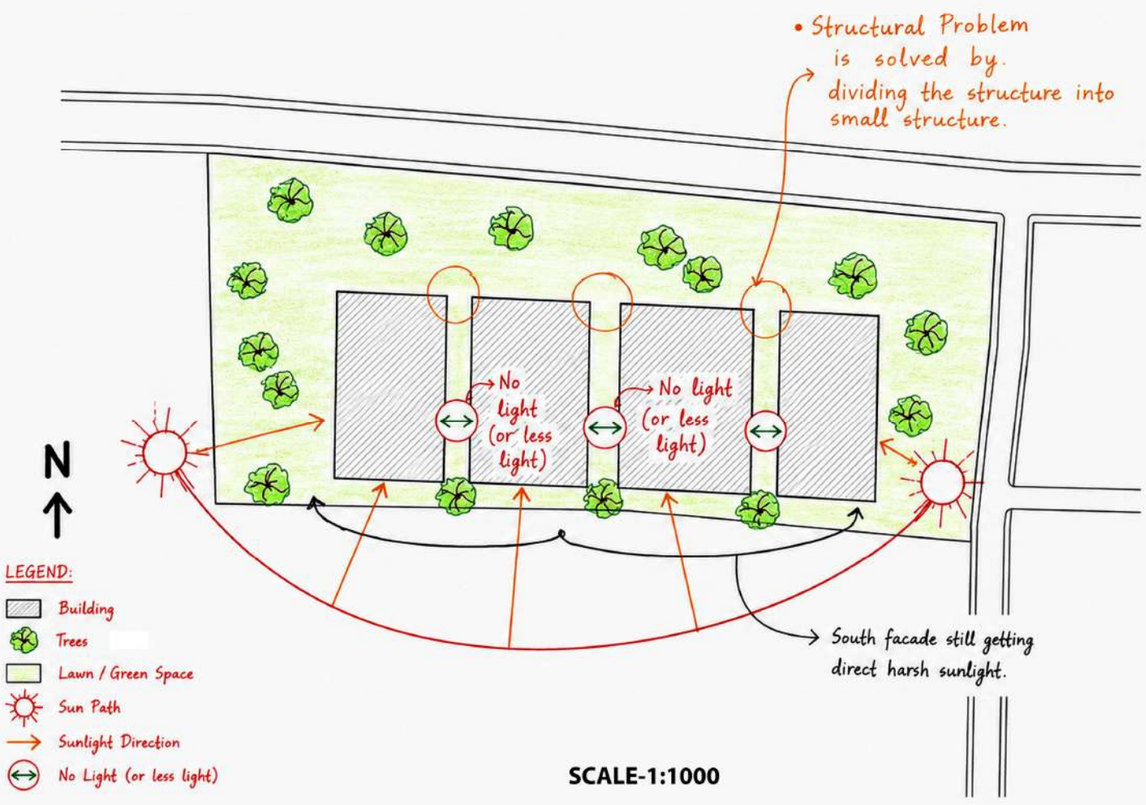
SCALE-1:1000

STEP:1



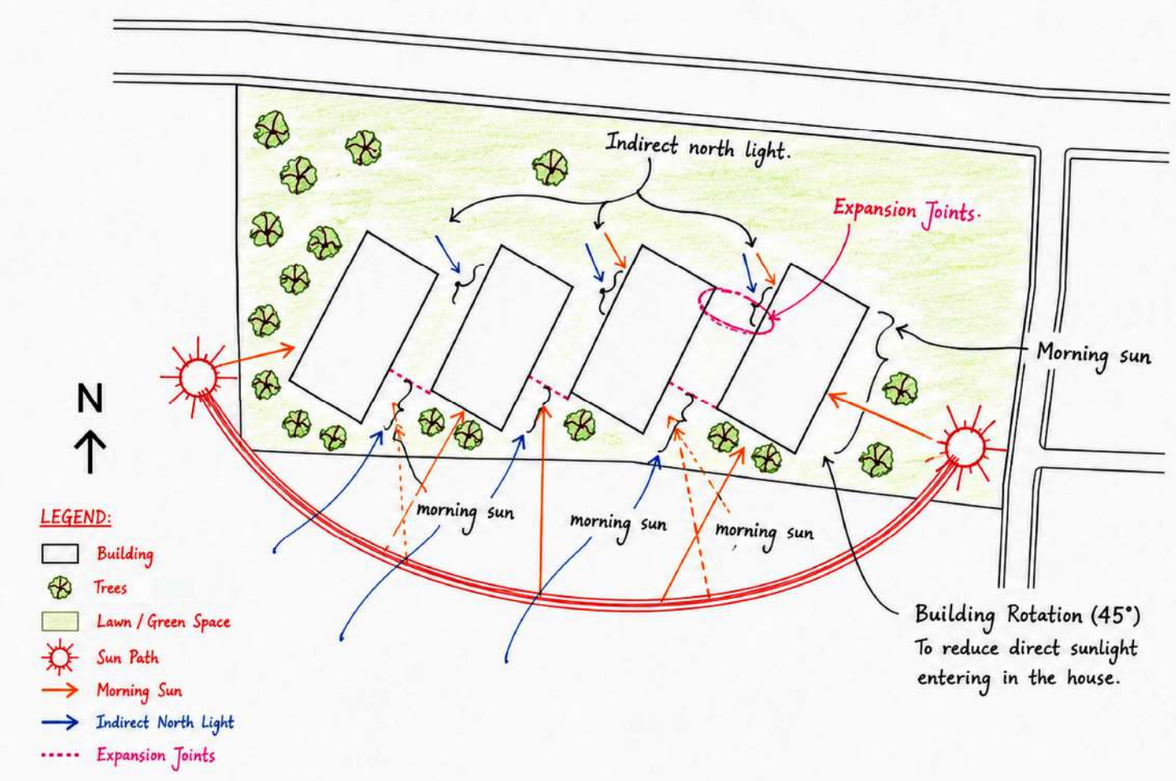
SCALE-1:1000

STEP:3



SCALE-1:1000

STEP:4



SCALE-1:1000

AREA STATEMENT

SITE AREA: 2acres (8094 sqm)
GROUND COVERAGE: 40% (3240 sqm)
FAR: 2
BUILDING HEIGHT: G+5
TOTAL BUILTUP: 16188sqm

SPACE	PER PERSON AREA sqm	AREA OF ONE UNIT sqm	QUANTITY	TOTAL AREA sqm
RESIDENTIAL				
HOSTLE OFFICE	_	35-40	1	40
SINGLE SHARING ROOM	_	25-30	20	600
DOUBLE SHARING ROOM	_	25-30	65	1965
TRIPLE SHARING ROOM	_	35-40	50	2000
ELECTRICAL ROOM	_	5	10	100
CANTEEN SERVICE AREA				
STAFF SHARED BEDROOM	06_09	43	1	43
WASHROOM	_	20	1	20
DRY STORAGE	_	18	2	36
COLD STORAGE	_	7	2	14
LAUNDRY	_	50	1	50
WASHED CLOTHES STORAGE	_	8	1	8
UNWASHED SLOTHES STORAGE	_	8	1	8
ELECTRICAL ROOM	_	6	3	18
KITCHEN AREA	_	120	2	240
TERRACE ACTIVITY SPACE	_	300	1	300
COMMERCE SPACE				
MINI SUPER MARKET	_	55	1	55
MEDICAL SHOP	_	30	1	30
CYBER+PRINT SHOP	_	30	1	30
FOOD SHOP	_	30	1	30
TEA+COFFEE SHOP	_	30	1	30
OUTDOOR SEATING	_	50	1	50
INDOOR SEATING	_	140	1	140
TOILET	_	45_50	1	50
STORAGE AND WASHING AREA	_	35_40	1	40
OUTDOOR SERVICE				
GENERATOR ROOM	_	10	1	10
SECURITY ROOM	_	6	3	20
			35% CIRCULATION	2074.45
			TOTAL:	8001.45

PARKING CALCULATION:

1 TWO-WHEELER PER 2 STUDENTS
BICYCLE PARKING FOR 30% STUDENTS
100 STUDENTS ÷ 5 = 20 BICYCLES

1 BICYCLE PER 5 STUDENTS
TWO-WHEELER PARKING FOR 50% STUDENTS
150 STUDENTS ÷ 2 = 75 TWO-WHEELERS

1 CAR PER 10 STUDENTS
CAR PARKING FOR 30% STUDENTS
100 STUDENTS ÷ 10 = 10 CARS

6 STAFF CAR PARKING
5% OF TOTAL STUDENTS

PARKING				
PARKING TYPE	**PER VEHICLE AREA sqm**	**NUMBER**		**TOTAL AREA sqm**
TWO-WHEELERS	2.5_3	75		200
BICYCLES	1.5_2	20		40
STUDENT CARS	12_20	10		200
STAFF CARS	12_20	6		120
			50% CIRCULATION	280
			TOTAL:	840

GOVERNMENT POLICY INTEGRATION

SOLAR SUBSIDY INTEGRATION:

- ROOFTOP SOLAR PANELS REDUCE DEPENDENCY ON CONVENTIONAL GRID ELECTRICITY.
- SOLAR-GENERATED POWER SUPPORTS COMMON AREA LIGHTING, LIFTS, CAFETERIAS, AND STREET LIGHTING.
- GOVERNMENT SUBSIDY SCHEMES IMPROVE ECONOMIC FEASIBILITY OF RENEWABLE INFRASTRUCTURE.
- NET-METERING ALLOWS EXCESS ELECTRICITY EXPORT BACK TO THE GRID.
- REDUCES LONG-TERM OPERATIONAL COST FOR RESIDENTS.

GOVERNMENT SCHEMES:

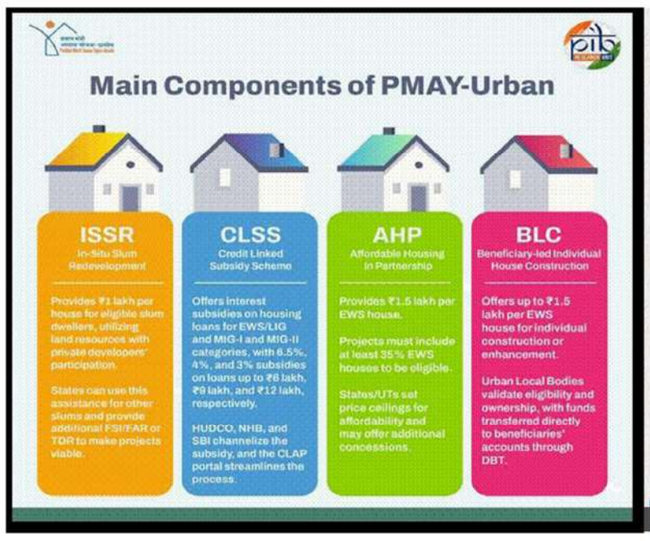
1. **PM SURYA GHAR MUFT YOJANA**
2. **KARNATAKA SOLAR ROOFTOP POLICY**
3. **MNRE SOLAR SUBSIDY SCHEME**





PRADHAN MANTRI AWAS YOJANA – URBAN (PMAY-U):

- SUPPORTS AFFORDABLE URBAN HOUSING DEVELOPMENT.
- ENCOURAGES HIGH-DENSITY MIXED-USE URBAN GROWTH.
- PROMOTES HOUSING NEAR WORKPLACES AND TRANSIT.
- **SUPPORTS EWS, LIG, AND MIG HOUSING CATEGORIES.**
- ENCOURAGES SUSTAINABLE INFRASTRUCTURE INTEGRATION.

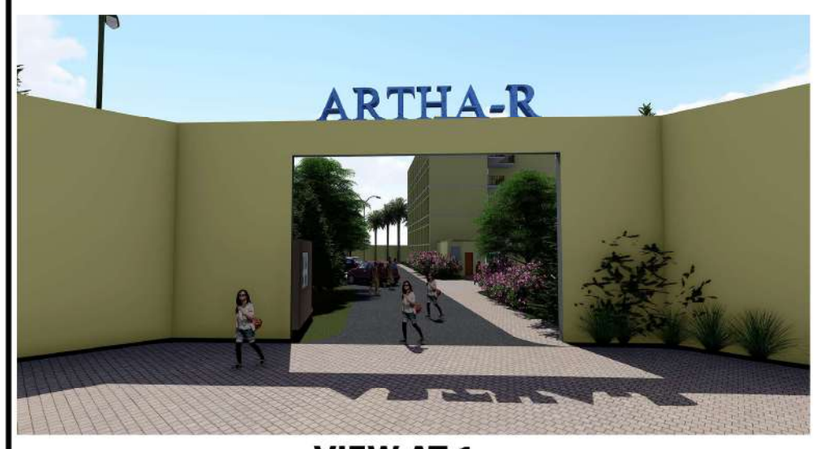




AFFORDABLE RENTAL HOUSING COMPLEXES (ARHC):

- ARHC SUPPORTS AFFORDABLE RENTAL HOUSING NEAR WORKPLACES AND INSTITUTIONS.
- REDUCES COMMUTING DISTANCE AND TRANSPORTATION DEPENDENCY.
- ENCOURAGES COMPACT WALKABLE COMMUNITIES.
- IMPROVES ACCESS TO DIGNIFIED URBAN LIVING.

SITE PLAN



VIEW AT 1



VIEW AT 2



VIEW AT 3



VIEW AT 4



VIEW AT 5



VIEW AT 6



SITE SECTION

SCALE-1:300



LEGEND:

- BOYS RESIDENCE
- GIRLS RESIDENCE
- SERVICE + CANTEEN
- COMMERCIAL SPACE
- ADMIN OFFICE

SITE PLAN

SCALE-1:300



FLOOR PLANS



LEGEND:

1. SINGLE ROOM
2. DOUBLE-SHARING ROOM
3. TRIPLE SHARING ROOM
4. LLAUNDRY ROOM
5. WASHED CLOTHES STORAGE
6. UNWASHED CLOTHES STORAGE
7. TOILET FOR STAFF
8. WASH AREA
9. SHARED ROOM FOR STAFF
10. DRY STORAGE 1
11. DRY STORAGE 2
12. ELECTRICAL ROOM
13. CANTEEN OPEN SEATING
14. CANTEEN INDOOR SEATING
15. FOOD COUNTER
16. HANDWASH
17. UTENSIL WASH AREA
18. DRY WASH AREA
19. KITCHRN
20. COLD STORAGE
21. TERRACE FOR ACTIVITY
22. TOILET
23. COMMON STORAGE

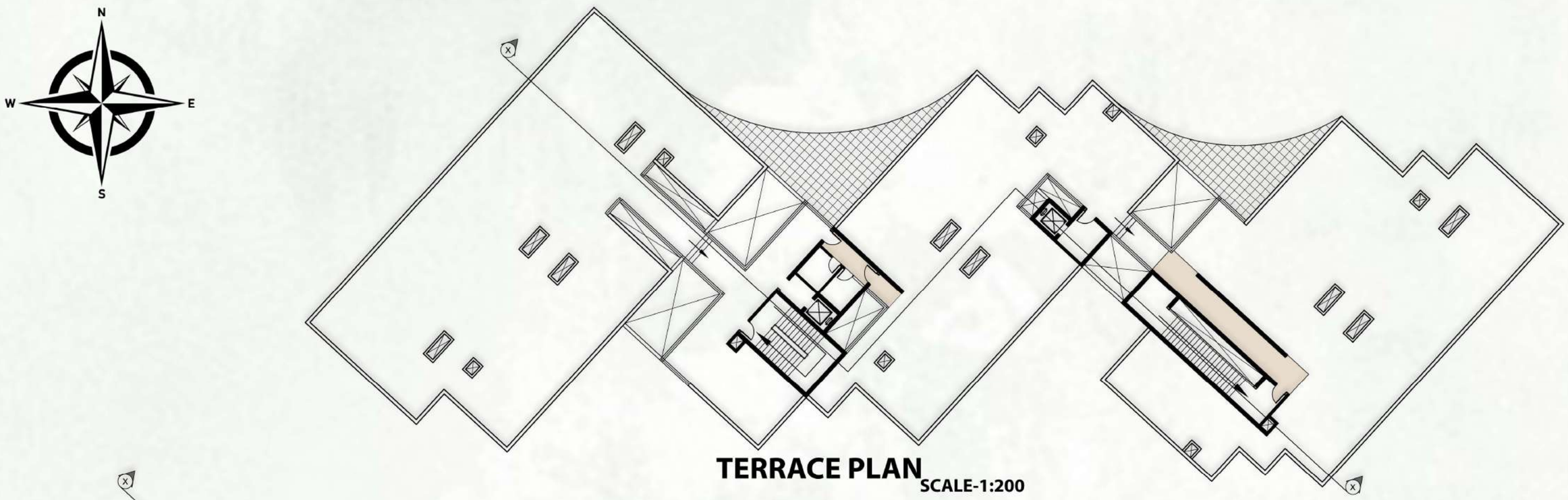
PLANS AND SECTION



VIEW FACADE VIEW



SECTION AT XX
SCALE-1:200



TERRACE PLAN
SCALE-1:200



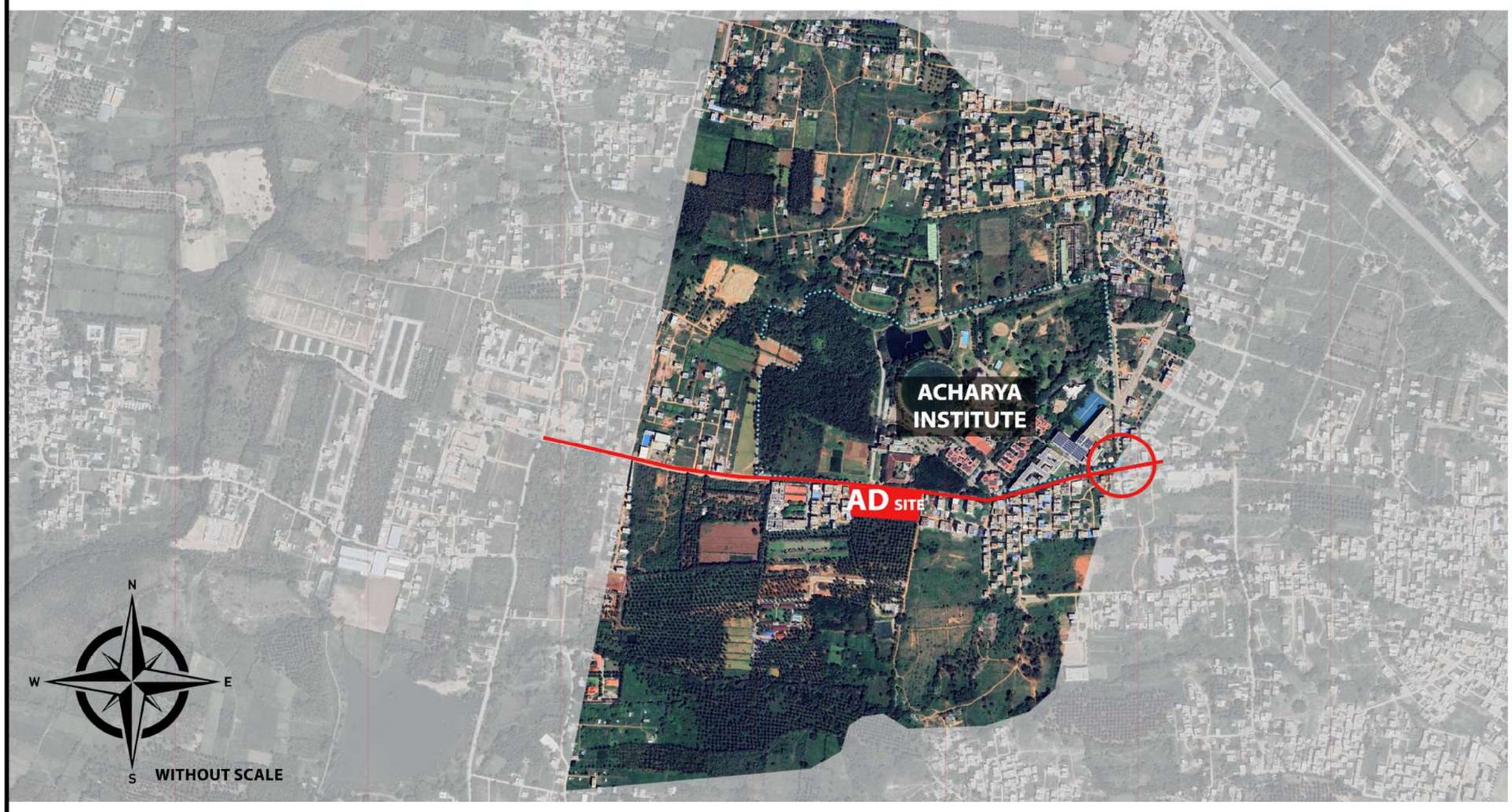
LEGEND:
1. SINGLE ROOM
2. DOUBLE-SHARING ROOM
3. TRIPLE SHARING ROOM

FOURTH FOOR PLAN
SCALE-1:200

ACHARYA FRONT ROAD

INTRODUCTION:

THIS IS ACHARYA MAIN ROAD, AN IMPORTANT CONNECTING ROAD USED BY STUDENTS, PEDESTRIANS, AUTO-RICKSHAWS, AND DAILY COMMUTERS. DUE TO UNORGANIZED ROADSIDE PARKING, LACK OF PEDESTRIAN INFRASTRUCTURE, AND ABSENCE OF ZEBRA CROSSINGS, THE ROAD EXPERIENCES TRAFFIC CONGESTION, PEDESTRIAN INCONVENIENCE, AND ACCIDENT RISK.



EXISTING CONDITION:



ISSUES IDENTIFIED:

- AUTO-RICKSHAWS WAITING NEAR THE MAIN CARRIAGEWAY CREATE TRAFFIC CONFLICTS.
- LACK OF ZEBRA CROSSING INCREASES PEDESTRIAN SAFETY RISK.
- NO PROPER FOOTPATH FORCES PEDESTRIANS TO WALK ON THE ROADWAY.
- IMPROPER ROADSIDE PARKING CAUSES VISUAL OBSTRUCTION FOR MOVING VEHICLES.
- TRAFFIC CONGESTION OCCURS DUE TO UNREGULATED STOPPING OF AUTOS.
- CONFLICT POINTS BETWEEN VEHICLES AND PEDESTRIANS INCREASE ACCIDENT POSSIBILITIES.





DESIGN:

- ZEBRA CROSSING PROVIDED FOR SAFE PEDESTRIAN MOVEMENT.
- DEDICATED RIGHT-TURN LANE IMPROVES TRAFFIC CIRCULATION.
- FOOTPATHS ADDED ON BOTH SIDES FOR PEDESTRIAN SAFETY.
- GREEN BUFFERS INTEGRATED TO ENHANCE STREETSCAPE QUALITY.
- TRAFFIC CALMING INTRODUCED TO REDUCE VEHICLE SPEED.
- PROPER STREET LIGHTING IMPROVES NIGHTTIME VISIBILITY.
- WELL-DEFINED ROAD EDGES PREVENT RANDOM PARKING.
- PEDESTRIAN SIGNAGE INCREASES DRIVER AWARENESS AND SAFETY.
- OVERALL DESIGN IMPROVES SAFETY, ACCESSIBILITY, AND TRAFFIC MANAGEMENT.

